

MACHINE LEARNING APPLICATIONS IN FINANCIAL LEASING

Group Id - 25-26J-179

Project Proposal Report

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B.Sc. (Hons) Degree in Information Technology

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Declaration

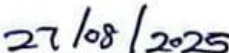
Declaration

I declare that this is my own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

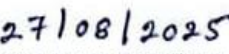
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The above candidates are carrying out research for the undergraduate dissertation under my supervision.


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Date

Abstract

The financial leasing sector is important to the economic growth by providing businesses and individuals with access to capital. However, as the financial leasing sector grows and develops, management of branch networks in a productive and efficient manner has become critical in remaining competitive and sustainable. Nevertheless, with increasing customer expectations and disparate market environments, traditional methods of evaluating branches are often too limited and cannot provide timely or accurate insights of branch operations. This implies the crucial need for sophisticated analytical methods adept at discovering substantial patterns in operational data to inform strategic decisions.

The Branch Performance Analysis module was developed to facilitate this type of evaluation, using clustering and aggregation methods to systematically evaluate data on a branch level. With clustering and aggregation everybody benefits by making the initial pile of branch data manageable, allowing clusters of branches with similar performance to be grouped together, which makes it easier to identify high performing branches that can be used as benchmarks and low performing branches requiring more targeted improvements. Further, the ability to track trends in efficiency data over time gives the organization a greater ability to deliver actionable data that informs resource allocation choices, maximizes productivity of scarce resources, and, enhance overall performance of the organization.

The goal of this innovation is to turn raw performance data into clear, actionable insights that allow financial institutions to take data-driven action. The Branch Performance Analysis component is hence solving the existing challenge of a lack of holistic or disconnected performance evaluation approaches by providing a systematic, analytical, and scalable performance monitoring approach. Overall, it adds to improved operational decision making, optimized resource allocation, and overall long-term sustainability of financial leasing institutions.

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List Of Abbreviations

Table 1- List of Abbreviations

Abbreviation	Description
BPA	Branch Performance Analysis
UI	User Interface
DB	Database
API	Application Programming Interface
SQL	Structured Query Language
ML	Machine Learning
KNN	K-Nearest Neighbors
CSV	Comma-Separated Values
GUI	Graphical User Interface
KPI	Key Performance Indicator

1. Introduction

Branch Performance Analysis (BPA) is gaining traction as a practical avenue for promoting efficiencies and competitiveness within financial institutions [1]. Evaluating the efficiency of branch operations has traditionally involved static reports, limited manual comparisons and assessments not always identifying trends, assessing variations in performance across each unique mechanism or branch, a set of insights that ultimately slows decision-making by leaving managers with little timely knowledge about each branch's strengths and weaknesses. BPA mitigates these decision-making limitations by harnessing data-driven techniques to reveal trends and analyze patterns that are not always visible by simply relying on standard reports [2].

BPA allows financial institutions to systematically analyze significant amounts of operational and transactional data by utilizing clustering and aggregation algorithms for analysis for unique performance characteristics [3]. When analyzing large numbers of branch operations and outputs, BPA groups branches that exhibit some similar performance characteristics for settings the framework of high-performing branches as a level of performance and those that have underperformed that can assist in designing corrective action [4]. This analytical application of BPA develops a higher level of transparency, and ultimately improves the decision framework on staff training, resource allocation and strategic planning.

The importance of BPA in the financial leasing sector is evident as institutions continually try to be more efficient while also responding to the increasing demands of new customer experiences [5]. In a world where operations excellence and customer experience are mutually dependent, being able to know with accuracy and timeliness how branches are performing will directly impact profitability and long-term viability. Continuous analysis of trends towards efficiency over time with BPA allows organizations to put into place more focused improvement programs and limit operational risks, meaning increasing overall competitiveness [6].

The implementation of BPA under financial systems also corrects the weaknesses of traditional performance evaluation. Unlike standardized performance review and reporting, BPA allows institutions to have dynamic, data driven intelligence that are not stagnant or traditionally aligned — allowing institutions at all levels to be adaptable to the changing business environment [7]. Thus, the Branch Performance Analysis component is not simply a monitoring tool; rather, it is a

strategic approach to enable innovation, rationalize the use of resources, and sustainably grow the financial leasing industry [8].

1.1 Background & Literature survey

Branch Performance Analysis (BPA) is a recent methodology that is becoming popular across financial institutions to improve operational efficiency and competitiveness. When evaluating branch performance, organizations traditionally relied on static reports, simple comparisons, or manual calculations to assess performance without understanding branch performance differences and trends. The limited capability of traditional branch performance analysis slowed decision making and limited management's understanding of the strengths and weaknesses in branch operations. By utilizing data driven methods, BPA presents an organized way, with analytical rigour, for organizations to find untapped relationships and trends that are not readily visible to traditional branch performance measurement process [9].

The expanding transactional and operational data available to branches provides organizations the opportunity to investigate performance differences across branches using clustering and aggregation algorithms. These clustering and aggregation algorithms help organizations identify groups of branch performance similarities. By identifying clusters, organizations will be able to identify high performing branches to act benchmarking examples, while also identifying low performing branches requiring corrective actions or additional resources. In doing so BPA increases transparency and supports well-informed decision making regarding resource allocation, staffing, operations and training and development, and strategic organizational planning. This analytical framework affords organizations the ability to look and act beyond a historical performance based view in searching for identifiable efficiencies and/or optimize organizational operations [10] [11].

The significance of BPA is most apparent in financial leasing because branch operations affect customer satisfaction and institutional profitability. Effective and timely information of branch performance enables an institution to be aware of how branch operational strategies align with customer expectations, regulations, and competitive pressures. BPA makes it possible to sustain

performance monitoring over time and thus develop improvement agendas, mitigate risk, and optimize operational efficiency [11].

Further, the application of BPA used in financial systems has the capability to change the traditional landscape of performance assessment. BPA replaces the static report with dynamic real-time intelligence that adapts as the business environment changes. This adaption ensures that timely clear actions are executed applying evidence-based decision making, strategic interventions, and intensity of resource utilization. Accordingly, BPA is not just an operational assessment tool, it is a catalyst for innovation that accelerates sustainable growth while driving competitiveness in financial leasing [11].

1.2 Research Gap

These are indicated in the following table where one is asked to show whether various BPA functions do or do not have indicated performance indicators, the type of gap that creates, and what is needed to fill that gap. The five essential attributes examine: using data-driven analytic methods to improve branch performance; real-time capacity for performance assessment; identification of which branches and sites are high- versus low-performing; support operational decision-making and resource allocation; and strategic integration with organization-wide planning and strategy. Across each item, check marks are displayed noting whether or not the item has been addressed. A ‘✓’ suggests that specific item has been covered in previous literature aside from “X” indicating that it does not address the item. From this table, one can know what gaps exist in the literature review and potentially how those gaps can be filled for the proposed recommendations.

Feature Analysis:

Performing data-based analysis of branch performance is rarely undertaken and some approaches rely on static reporting or just a comparison and therefore reveal the literature's coverage of this important feature is likely partial [1] [3]. It is essential for the framework's development to address all the components of data-based analysis followed by a more complete and consistent view of

branch performance. Some documents' exclusion of the feature suggests the need for a consistent approach that fully utilizes data that can be transformed into actionable knowledge.

In terms of real-time updates of branch performance, such features are large absent from studies identified, most only considered it an ancillary activity [4], resulting in a significant gap, as real-time integration to performance monitoring will be critical in adaptive or responsive management strategies. The proposal aims to fill a sizeable gap in that real-time updates should be considered an essential requirement, for timely and informed decision-making.

While some literature indicate some awareness of high- and low-performing branches, there is little consistent methodology [1] [3]. This is critical so that high-performing branches can be benchmarks and low-performing branches can be targeted for interventions. The proposed solution offers standardized methodologies for the categorization of branch performance.

Support for resource allocation and operational decision making is absent in many approaches to performance analysis [4] [7], leaving a fissure preventing analysis to be completed to produce actionable management plans. By including this feature, the proposed framework provides a way to incorporate BPA insight as the foundational decision framework for resource allocation and operational planning at the branch staff level.

Linkage to strategic organization planning is complex and not often discussed, meaning there is little intent to connect branch performance understanding to organization end goals [3]. The solution being proposed proposes to tackle this by utilizing operational improvements and decisions at the branch level to facilitate competitively sustainable growth.

Overall, it is possible to assess by reviewing the table what aspects are covered adequately and what is absent in order to enact effective branch performance analysis. The absence of these aspects highlights the need for the proposal solution as it is attempting to fill these voids in order to enhance the overall effectiveness of branch performance analysis in financial institutions.

Table 2- Research gap

Feature / Criteria	Reference [1]	Reference [3]	Reference [4]	Reference [7]	Proposed Solution
Using data-driven analytic methods to improve branch performance	✓	✓	X	X	✓
Real-time capacity for performance assessment	X	X	✓	X	✓
Identification of high- vs low-performing branches	✓	✓	X	X	✓
Support operational decision-making and resource allocation	X	X	✓	✓	✓
Strategic integration with organization-wide planning and strategy	X	✓	X	X	✓

1.3 Research problem

The financial services sector is an area of great potential for economic development in Sri Lanka, and financial leasing and the access to capital that leasing can provide to both businesses and individuals continues to be a critical element. Unfortunately, the financial services sector in Sri Lanka has a significant shortcoming - the inability to provide good practices to continuously gauge and track performance on an operational level of leasing branches over time [1]. Current industry practices often consist of limited key performance measures, static reports or manual comparisons, providing only a snapshot view of performance which is neither timely nor complete for decision-making purposes [11]. Therefore in many instances, the leaders of important business segments, and decision-makers, have to make decisions with incomplete and/or old data resulting in inefficiencies, mismatched strategies and lost opportunities for growth. This in turn leaves

underperforming branches ignored, but equally high-performing branches are taken for granted in addition to not being used sufficiently as a model to other branches [3].

To help solve this vital question, this study proposes a dedicated Branch Performance Analysis (BPA) component, which is intended to regularly track and analyze efficiency of the financial leasing branch operations. Utilizing clustering and aggregation algorithms, BPA will be able to identify patterns and cluster branches based on performance similarities to uncover both strengths and weaknesses across the network [1] [3]. This development will add to the evidence-base for using high-performing branches as exemplar models for improvement, and applying interventions for development outcomes to low-performing branches [3].

The planned BPA component will also facilitate branch-level management and can serve as a strategic planning tool for the organization. This will allow for effective resource allocation, better informed decision-making, and strategic performance enhancement initiatives that are aligned with the longer term institutional objectives [11]. In summary, the implementation of a BPA system will change how we think about and evaluate branch performance and will reduce the time and distance needed to integrate the disjointed ways of thinking about performance with the need for complete and decision-sensitive evaluation. Ultimately, this will contribute to the adaptability, competitiveness, efficiency, and sustainability of financial institutions, in an ever-changing financial services environment [1] [3].

2. Objectives

2.1 Main objective

The proposed project will fundamentally reform how financial leasing branch operations are managed and evaluated. By incorporating modern data driven techniques, such as clustering algorithms, aggregation algorithms, and performance analytics, the analytical tools proposed are designed to improve organizational awareness and operational information is all about identifying patterns, grouping branches given the performance data, and discerning clear distinctions about what makes a good branch and what makes a poorly performing branch. The aim of the proposal is to remove some of the potential errors made in traditional performance of a branch context, adopting a more cohesive approach relating to

groups of branch data and putting data in timely context that allows informed decisions to be made eliminating the guessing of strategic planning and resource allocation and better the institutions overall effectiveness. The ultimate goal of the project is to create a more informed financial leasing network that can rapidly adapt to changing business environments and improve on institutional effectiveness.

2.2 Specific objectives

The BPA component will be implemented to provide continuous monitoring of financial leasing branch's operational efficiency over time. This will provide a complete and contemporary view of branch performance, which goes beyond static reports or manual comparisons of branch performance, and aids management in making more informed decisions.

Clustering and aggregation algorithms will be used to detect patterns of branch performance and categorize branches according to similarities in the performance metrics. This way, both branch performance high-pots can be identified for use as models, and low-performers can be targeted for operational corrections. These activities will increase and involve a higher base with respect to operational efficiency.

Integration of real-time updates into the BPA will be available so that branch managers and decision-makers can actively receive timely insights and suggestions for action to take back to performance and operationally relevant information sets. This encourages operational decisions and strategic planning, as well as, reflects adaptive and responsive management strategies.

This BPA component will also have an interactive and user-friendly interface that accurately describes the data, provides performance reports and analytical tools and displays intuitive data visualizations. The interface will expedite how management understands branch performance and how they act to increase operational efficiency and service quality.

Ultimately, the BPA component's goal is to present opportunities to develop a structure for effectively using branch performance insights for resource allocation, as well as informing branch performance improvement programs and organization-wide planning. This will lead to improved competitiveness, operational effectiveness and overall sustainability across the financial leasing network, using an evidence-based approach.

3. Methodology

3.1 System Diagram

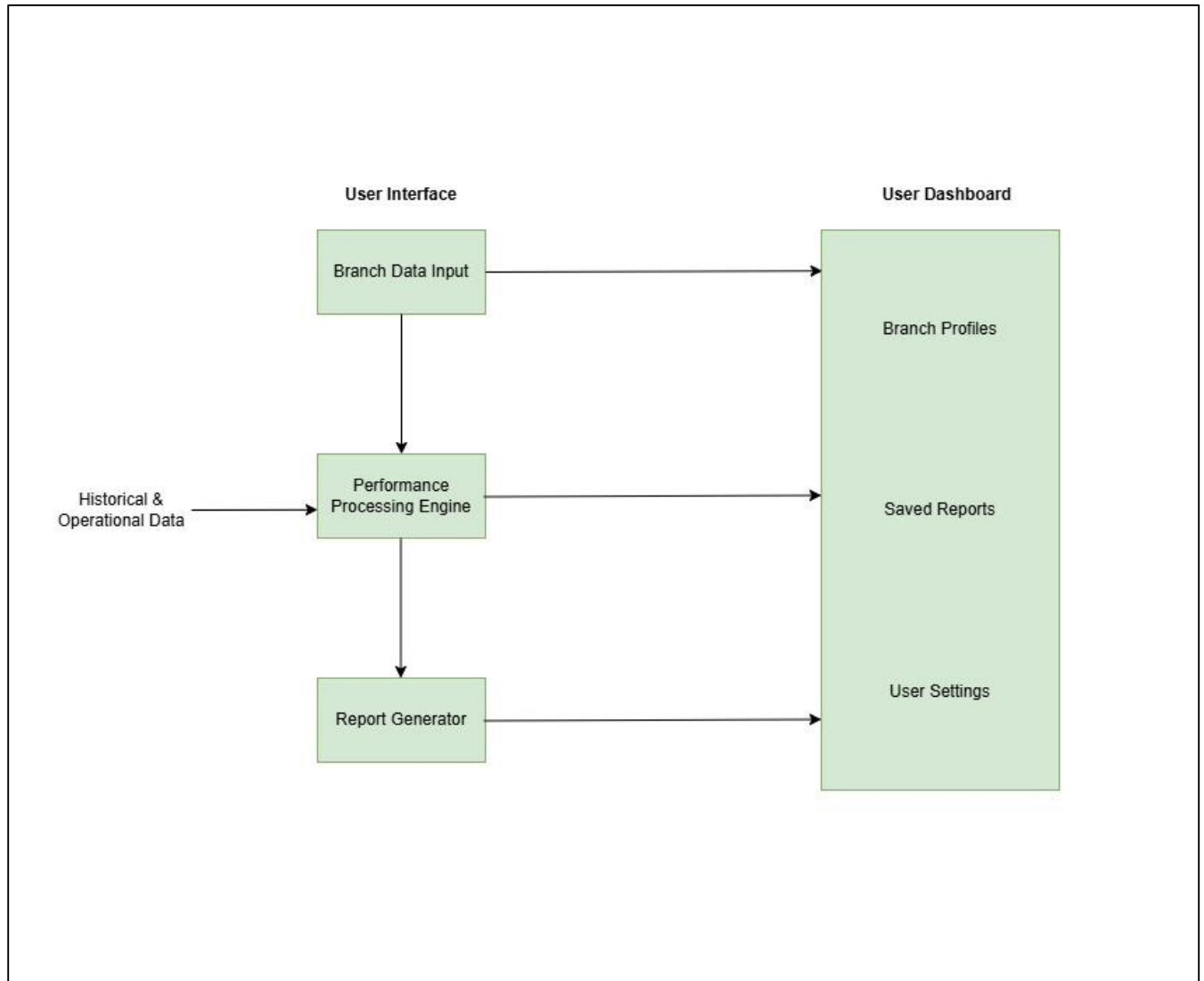


Figure 1- System Diagram

The system diagram depicts the organization and data flow for the Branch Performance Analysis system. This feature is designed to offer data-based insights on the operational health and productivity of distinct branches, so management can make data-driven strategic decisions. The following outlines the different elements and their relationships.

User Interface:

- **Branch Data Input:**

This is the first stage in the analytical workflow. This is a distinct interface for uploading, validating and submitting a new operational and financial data set from one of the branches. Once the branch data set has been submitted, it begins the whole performance analysis system by sending or delivering the new data set to the back-end Performance Processing Engine.

- **Performance Engine:**

This sub-component provides a visual control panel and status view of the analysis engine itself. Here users can see the status of processing data submissions, determine which analytical models (or clustering algorithms) they would like to run (e.g., Group by Region, or Identify Top Performers), and see a preliminary result before generating a formal report. The performance engine acts as a bridge between raw branch data input, and generating the final report as the end product from the system.

- **Report Generator:**

The Report Generator interface gives users the option to configure and request a final detailed performance report, as well as view the report directly. The user will identify which branches within the analysis to use, the date range for the report and how to display the selected calculated KPIs. The Report Generator takes processed data from the Performance Processing Engine and outputs the data in a cleaned, consumable form through charts, graphs, and tables suitable for export.

User Dashboard:

- **Branch Profiles:**

This is a user-generated list of chosen “favorite” or otherwise frequently monitored branches. Rather than needing to choose from the entire branch list each time the user wants to run a report, they will save a number of individual branches here, for easy access and reporting within the Report Generator.

- Saved Reports:

Enables users to save fully configured reports in the Report Generator so that they can easily access them in the future. Users can one-click to view reports that are scheduled regularly (e.g., "Weekly Performance Snapshot") without re-configuring their parameters.

- User Settings:

This element accounts for each user's preferences and defaults (e.g., default home branch, default date range (e.g., "Last Quarter"), and colors of chart themes). Preferences personalize the default behavior of the Report Generator to the extent it may do so for other User Interface (UI) components.

3.2 Work breakdown structure

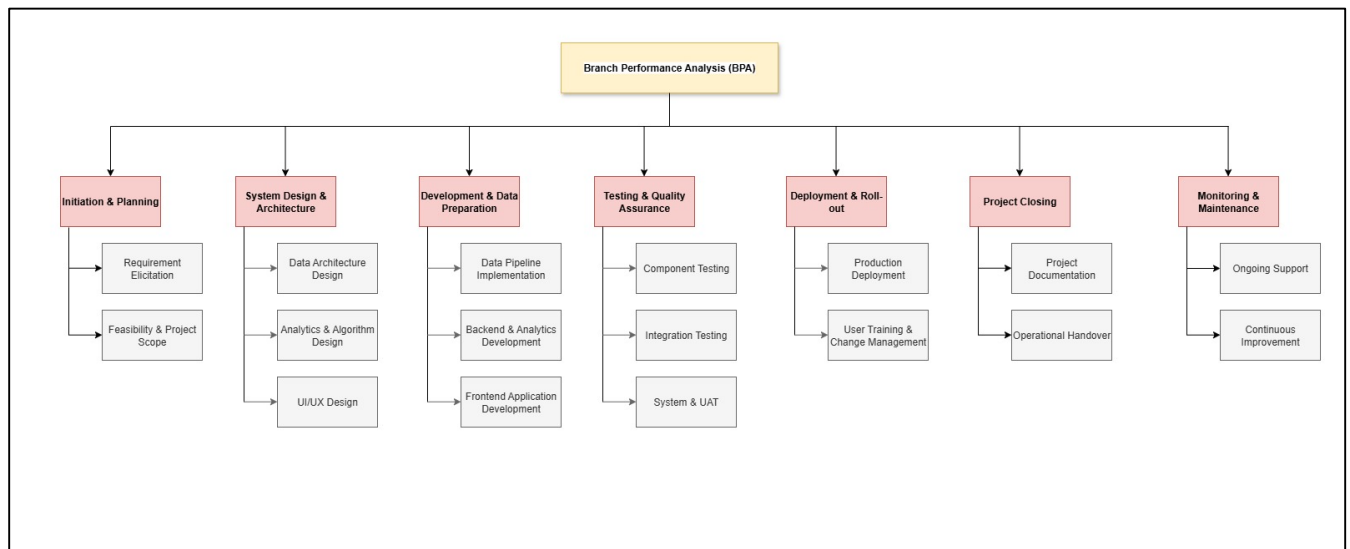


Figure 2 – Work breakdown Structure

3.3 Gantt Chart

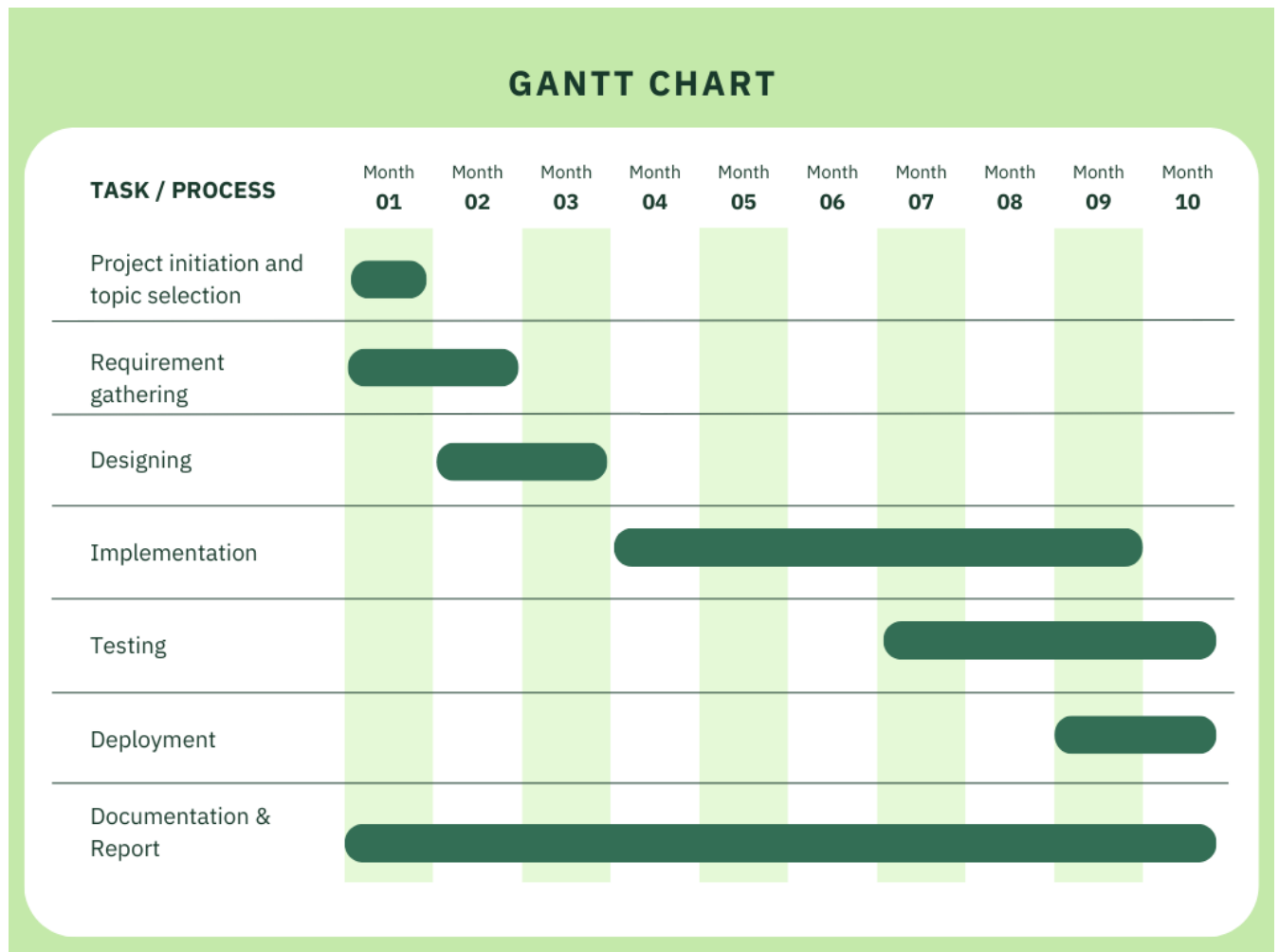


Figure 3- Gantt Chart

4. Project Requirements

4.1 Functional Requirements

- **Branch Data Collection**

Collect operational, transactional, and performance data from every branch to capture a whole picture.

- **Performance Dashboard**

Create an interactive dashboard with performance trends, allowing management to view comparative branch performance, best and worst performing branch highlights.

- **Reporting & Alerts**

Provide automated reports about branch performance and alerts explaining when performance is deemed low compared to expectations/ benchmarks.

- **Historical & Trends Analysis**

Provide functionality for tracking branch performance over time down to the employee level so management can make better strategic decisions and monitor improvement programs.

- **Cross Platform Access**

Ensure the BPA Dashboard and tools can be accessed using web browsing, desktop, and mobile platforms for management viewing ease.

4.2 Nonfunctional requirements

- **Performance**

It is essential that the system quickly process large amounts of branch data and provides analytics in near real time to support timely decision making.

- **Scalability**

The system must be able to manage the growth of branches as well the growth in the amount of data as the institution grows.

- **Security**

Include appropriate authentication, encryption, and access controls to safeguard branch and customer sensitive information.

- **Usability**

The system should have an intuitive interface allowing Regional and District Managers with a limited technical background to easily interrogate and synthesis data.

- **Reliability**

The system must deliver operational availability and continuous data integrity with limited to no down time and availability across multiple platforms.

- **Maintainability**

The ongoing development and expansion of appropriate Key Performance Indicators, Metrics, Analysis, and reporting features should be easily accommodate.

4.3 User Requirements

- **Ease of Access**

Managers should access the BPA dashboard and analytics easily from a web browser or mobile devices.

- **Performance Insights**

Users should be able to readily view summaries of branch performance and identify trends.

- **Comparisons & Benchmarking**

Users should be able to compare branches to each other, but also compare branches against the best-performing clusters.

- **Cross-Platform Compatibility**

The BPA tools should seamlessly function on desktops, tablets and mobile devices to be used by managers in the field or at headquarters.

4.4 System requirements

Tools and Techniques

Tools & Technologies	Purpose
Frontend Development(React)	Building interactive user interfaces and dashboards for leasing data analysis
HTML5, CSS3, TypeScript	Structuring, styling, and scripting the frontend application
Backend Development(.NET Core)	Developing APIs for data access, processing, and integration with ML models
REST API	Communication between frontend and backend
Machine Learning & Data Science(Python (scikit-learn, TensorFlow, PyTorch))	Data preprocessing, feature extraction, model training and evaluation
Pandas, NumPy	Data handling, manipulation, and analysis
Matplotlib, Seaborn	Visualization of data trends and model results
Database Management(SQL Sever / MY SQL)	Storage and querying of leasing datasets
Project Management(Jira)	Task tracking, sprint planning, and team collaboration
Version Controlling(Git & GitHub)	Version control and collaborative development
Collaboration and Communication(Microsoft Teams)	Real-time communication and project discussions
Compliance & Standards	Ensuring compliance with financial reporting standardss
Reporting & Documentation(Ms Word)	Preparing reports, documentation, and data presentation

Table 3- Tools and Techniques

5. Budget and budget justification

Category	Estimated Cost	Justification
Computing Resources (Local / University Lab)	free	Use existing university lab PCs/servers; no extra cost
Software & Tools	free	Open-source tools (Python, Scikit-Learn, TensorFlow, Pandas, NumPy, Matplotlib).
Data Preprocessing	free	Research assistant/student effort to clean 180,000+ rows and 10+ columns.
Model Development & Testing	Rs.5000	Effort to develop, tune, and validate ML models for impairment prediction
Statistical & Financial Analysis	free	Exploratory data analysis, IFRS 9 alignment, and feature selection.
Dissemination (Reports / Printing)	Rs.1500	Printing, report/thesis preparation, minor publication costs
Contingency (10%)	500	For any minor unforeseen research-related costs

Table 4- Budget and budget justification

6. References

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7. Appendices

- System Diagram

https://drive.google.com/file/d/16Xk9de-MJAT_LI3DB_8yNw_qXQryTnxT/view?usp=drive_link

- Work breakdown Structure

https://drive.google.com/file/d/1yuy12F18Y8S-ug07lOxq_5rekaDEJAKF/view?usp=drive_link

The screenshot shows the Turnitin interface. At the top, the Turnitin logo is on the left, and a navigation bar on the right contains links for 'Manuji Weerasinghe', 'User Info', 'Messages', 'Student', 'English', 'Community', 'Help', and 'Logout'. Below the logo, a 'Class Portfolio' tab is visible. The main content area has a breadcrumb trail: 'NOW VIEWING: HOME > RESEARCH PAPER CHECKING > RESEARCH PAPER CHECKING'. A section titled 'About this page' explains that this is the assignment dashboard for uploading submissions and viewing reports. Below this, a button labeled '> Research Paper Checking' with a help icon is shown. A table displays submission data:

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